

1 • Probability through an AI-assisted ticket-triage problem

Data Analytics for AI • Executive M.Tech • IIT Jammu
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Rigorous supplement

Purpose and pace

We begin with a six-ticket routing audit and use it to motivate the mathematical objects needed to describe events and assign probabilities. The sequence is:

1. Compare reviewer and model classifications, including a ticket with two required destinations.
2. Compare historical interpretations through the audit's agreement fraction.
3. Pose Bertrand's random-chord question and note its different answers under different sampling procedures.
4. Introduce Kolmogorov's set-based probability space, then specify uniform selection from the six tickets and define events as sets.
5. Introduce the triplet's requirements and a random variable.

The same six ticket IDs are used throughout. The meeting duration does not set a required pace; stop at a checkpoint and continue at the next meeting when needed.

1. A ticket-routing audit

An IT service desk is checking whether an LLM routes incoming tickets to the appropriate department. The pilot uses three reference outcomes:

- **Billing** when Billing is required;
- **Account Support** when Account Support is required;
- **Other** when neither Billing nor Account Support is required.

"Other" is the complement category for this two-destination question: the reviewer finds that neither Billing nor Account Support is required. Another team may handle such a ticket; "Other" does not mean that the ticket was unreviewed or its status is unknown. A ticket may require both Billing and Account Support, so the reference outcome can contain both labels.

The table is a hypothetical audit. Reviewer labels are operational reference decisions; they are not asserted to be error-free ground truth. For t_1 , the LLM reports $P(\text{Billing required} \mid x) = 0.55$ and $P(\text{Account Support required} \mid x) = 0.35$. To obtain the hard model classification shown in the table, apply a threshold of 0.50 separately to each destination. The other rows show the resulting model classifications.

Ticket	Request summary	Reviewer reference	Model classification
t_1	"I was charged twice, and now I cannot log in to my account."	Billing and Account Support	Billing
t_2	A duplicate charge appears on the monthly invoice	Billing	Billing
t_3	Cannot log in after changing the phone used for MFA	Account Support	Account Support
t_4	A delivery-status question unrelated to billing or account access	Other	Account Support
t_5	A subscription renewal was charged twice	Billing	Billing
t_6	Account locked after a password reset	Account Support	Account Support

The model and reviewer agree exactly on t_2, t_3, t_5, t_6 . They disagree on t_1 , where the model omits Account Support, and t_4 , where the model assigns Account Support but the reviewer assigns Other. Thus the observed agreement fraction is 4/6.

Ask:

What does 4/6 describe? Under what experiment would it be a probability, and what additional assumptions would connect it to future tickets?

The fraction is an observed summary of these six records. It becomes a probability for selecting uniformly from these six ticket IDs once that selection experiment is specified. It does not, by itself, describe future service-desk traffic.

Checkpoint 1. Identify the reference outcome for t_1 and explain why "Other" applies to t_4 but not to t_1 .

2. Historical interpretations of probability

Probability has been interpreted in several ways. The accounts below answer different questions about what the number 4/6 might mean. None follows just from seeing six rows in a table.

Classical equal-likelihood

The classical rule counts favourable outcomes among equally likely cases. If one ticket ID is selected uniformly from the six IDs, four selections produce agreement, so

$$P(\text{agreement}) = \frac{4}{6} = \frac{2}{3}.$$

This calculation is valid for that selection experiment. It does not imply that future tickets are equally likely to be any of these six records, or that the six records represent future traffic. A finite list defines possible cases; the uniform selection rule supplies the equal weights.

The partition chosen for counting also matters. Group the tickets by the pair (reviewer outcome, model classification):

Reviewer outcome	Model classification	Ticket IDs	Exact match?
Billing and Account Support	Billing	t_1	No
Billing only	Billing	t_2, t_5	Yes
Account Support only	Account Support	t_3, t_6	Yes
Other	Account Support	t_4	No

There are four occupied pair types in this audit. If an experiment selected one of these four types uniformly, two of the four would be exact matches, giving $2/4$. If it selected one of the six ticket IDs uniformly, the pair types would instead have probabilities $1/6, 2/6, 2/6, 1/6$, giving $4/6$. There are eight possible combinations in the full product of four reviewer outcome types and two model classifications, but only four occur in this table; selecting uniformly among all eight would be yet another experiment. Equal weighting of pair types does not follow from grouping tickets that way. Counting requires a stated set of cases and a reason to treat those cases as equally likely.

Frequency

The observed agreement proportion is

$$\widehat{p}_6 = \frac{1}{6} \sum_{i=1}^6 I_i = \frac{4}{6},$$

where $I_i = 1$ when the model classification exactly matches the reviewer's reference outcome for ticket t_i , and $I_i = 0$ otherwise. A frequency interpretation connects a probability to the limiting relative frequency in a specified repeatable process. This audit records one finite set of six tickets; it does not establish that limit or show that future ticket mix and model performance will remain the same.

Repeatedly sampling the LLM for the same text under randomized decoding would define another process. Its output frequency would describe generated classifications, not necessarily agreement with reviewer decisions. The process and event must be stated before interpreting a frequency.

Subjective probability and Bayesian updating

A subjective interpretation treats probability as a person's degree of belief given available information. An analyst might assess the chance that the model agrees with the review protocol on a future ticket, using this audit and other stated information. The six rows do not determine that assessment uniquely. Coherence constrains probability assignments; subjective does not mean arbitrary. Countable additivity is an additional condition in the Kolmogorov model and does not follow from every finite coherence argument.

Bayesian updating describes how an assessment changes when evidence is observed. If H is the proposition that the next ticket's model classification will agree with its reference outcome and D is additional evidence, Bayes' rule gives

$$P(H \mid D) = \frac{P(D \mid H)P(H)}{P(D)}, \quad P(D) > 0.$$

The formula does not supply the prior or likelihood; those come from a model and stated information. Conditional probability and Bayesian updating will be developed later.

Logical or evidential probability

A logical or evidential interpretation aims to quantify how strongly stated evidence supports a proposition. Here the evidence could include the six audit rows, the review protocol and assumptions about the future ticket population. A numerical answer requires a rule connecting that evidence and those assumptions to a degree of support. If the rule invokes indifference, it must specify which cases are treated equally and why. Counting ticket IDs and counting occupied label-pair types give different answers because they describe different selection rules. This identifies a problem with an unspecified indifference rule; it does not refute every logical account.

What the interpretations leave for a probability model

Classical probability emphasizes equal likelihood, frequency emphasizes repetition, subjective probability represents an assessment given information, and logical accounts specify evidential support. A propensity account would treat the configured randomized generation process as the chance-producing mechanism. These accounts differ in interpretation. A mathematical calculation also needs the possible outcomes, the event of interest and a probability assignment. The next example shows why the sampling procedure is part of that specification.

Checkpoint 2. For each interpretation, state what additional assumption or information is needed to use it for a claim about future tickets.

3. A second question: Bertrand's random chord

Draw an equilateral triangle inside a circle. Consider the question:

If a chord is selected at random from the circle, what is the probability that its length exceeds the side length of the triangle?

The phrase "selected at random" does not specify how the chord is generated. Bertrand described different sampling procedures for chords, which give $1/3$, $1/2$ and $1/4$ for the same length comparison. These values follow from different probability distributions induced by the respective sampling procedures; the problem statement alone does not say which procedure to use. John D. Norton's University of Pittsburgh page presents the three sampling arguments and their values.

We will not choose among the procedures or derive the answers in this section. The example identifies information a probability question can leave unspecified: the outcomes and, especially, the sampling rule that determines their probabilities.

Kolmogorov's set-based framework

Kolmogorov's axiomatic framework gives probability a mathematical foundation in terms of sets and a measure. Its three objects are written

$$(\Omega, \mathcal{F}, P).$$

Here Ω is the sample space of possible outcomes, $\mathcal{F}$ is the collection of events to which probabilities are assigned, and $P: \mathcal{F} \rightarrow [0, 1]$ is the probability measure. The event collection has closure requirements, and the measure obeys nonnegativity, normalization and countable additivity. We will define and apply each object using the ticket audit. The supplement returns to Bertrand's question and specifies the sampling models in full.

4. Define the experiment and its events

To calculate a probability for the audit, specify an experiment:

Select one of the six ticket IDs uniformly at random.

The ticket records, reviewer decisions and model classifications are fixed. The selected ticket ID is the outcome that varies. Thus the sample space is

$$\Omega = \{t_1, t_2, t_3, t_4, t_5, t_6\}.$$

Define events from the reviewer's reference classification:

$$B = \{t_1, t_2, t_5\}, \quad A = \{t_1, t_3, t_6\},$$

where B is the event "Billing is required" and A is the event "Account Support is required." The event that neither is required is

$$O = (B \cup A)^c = \{t_4\}.$$

In this model, O is precisely the "Other" reference outcome. It is disjoint from B and A . The ticket t_1 belongs to both B and A , because both destinations are required:

$$B \cap A = \{t_1\}, \quad B \cup A = \{t_1, t_2, t_3, t_5, t_6\}.$$

For sets $C, D \subseteq \Omega$:

$$C \cap D = \{t \in \Omega: t \in C \text{ and } t \in D\},$$

$$C \cup D = \{t \in \Omega: t \in C \text{ or } t \in D\},$$

$$C^c = \Omega \setminus C.$$

The "or" in a union is inclusive. The outcome t_1 is counted in both marginal events, so to count tickets requiring at least one named destination we use the disjoint pieces $B \setminus A$, $A \setminus B$ and $B \cap A$:

$$B \cup A = (B \setminus A) \cup (A \setminus B) \cup (B \cap A).$$

Under uniform selection, their probabilities are $2/6$, $2/6$ and $1/6$. Therefore

$$P(B \cup A) = \frac{2}{6} + \frac{2}{6} + \frac{1}{6} = \frac{5}{6}.$$

Equivalently,

$$P(B \cup A) = P(B) + P(A) - P(B \cap A) = \frac{3}{6} + \frac{3}{6} - \frac{1}{6} = \frac{5}{6}.$$

The complement is O , so $P(O) = 1/6$. The reviewer outcomes form four possible destination sets: both, Billing only, Account Support only, or neither. "Other" refers only to the last of these in this example.

Checkpoint 3. Give the ticket IDs in B , A , $B \cap A$, $B \cup A$ and O . Explain why the probability of the union is not the sum of the two marginal probabilities.

5. The probability space and its requirements

The experiment is represented by the probability triplet

$(\Omega, \mathcal{F}, P)$.

Object	Ticket-audit model	Meaning
Ω	$\{t_1, t_2, t_3, t_4, t_5, t_6\}$	Possible outcomes: ticket IDs that may be selected
$\mathcal{F}$	2^Ω	Events: all subsets of the six ticket IDs
P	$P(E) = E /6$ for $E \subseteq \Omega$	Probability measure under uniform selection

An outcome such as t_2 is an element of Ω . An event such as B is a subset of Ω . The measure P takes an event as input and returns a number in $[0, 1]$. Here the uniform rule assigns probability $1/6$ to each singleton outcome.

The event collection $\mathcal{F}$ must be a **sigma-algebra** on Ω . Its requirements are:

1. **Contains the sample space:** $\Omega \in \mathcal{F}$.
2. **Closed under complements:** if $E \in \mathcal{F}$, then $E^c = \Omega \setminus E \in \mathcal{F}$.
3. **Closed under countable unions:** if $E_1, E_2, \dots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} E_i \in \mathcal{F}$.

These conditions ensure that "not E " and "at least one of $E_1, E_2, \dots$ " are events whenever the component sets are events. Since this example has a finite sample space, taking $\mathcal{F} = 2^\Omega$, the power set, satisfies all three requirements.

The probability measure satisfies Kolmogorov's axioms:

1. **Nonnegativity:** $P(E) \geq 0$ for each $E \in \mathcal{F}$.
2. **Normalization:** $P(\Omega) = 1$.
3. **Countable additivity:** if $E_1, E_2, \dots$ are pairwise disjoint events, then

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i).$$

For the six singleton events, additivity gives $P(\Omega) = 6(1/6) = 1$. Also B and B^c are disjoint and exhaustive, and $P(B) + P(B^c) = 3/6 + 3/6 = 1$. Closure requirements apply to $\mathcal{F}$; the three axioms apply to P .

A probability measure is a set function from events to the unit interval. The event B is a set of tickets; $P(B) = 1/2$ is a number. This keeps the sample point, event and probability value distinct.

Checkpoint 4. State the sample space, event collection and probability measure. Verify nonnegativity, normalization and additivity for two disjoint events from the table.

6. A random variable for the same audit

The reviewer event B can be represented by a numerical function. Define the Billing indicator

$$X_B: \Omega \rightarrow \mathbb{R}, \quad X_B(t) = \begin{cases} 1, & t \in B, \\ 0, & t \notin B. \end{cases}$$

Its values are

$$X_B(t_1) = 1, X_B(t_2) = 1, X_B(t_3) = 0, X_B(t_4) = 0, X_B(t_5) = 1, X_B(t_6) = 0.$$

A random variable is a set function from the outcome space to the real line. Once the outcome is specified, the function's value is fixed. It is not a random mechanism and it is not the probability measure. In this experiment, the ticket ID is selected at random; X_B records whether its reviewer reference includes Billing.

Image: values produced by a set of outcomes

For $E \subseteq \Omega$, the image of E under X_B is

$$X_B(E) = \{X_B(t) : t \in E\}.$$

For the Account Support event $A = \{t_1, t_3, t_6\}$, t_1 also requires Billing, while t_3 and t_6 do not. Hence

$$X_B(A) = \{0, 1\}.$$

This is a set of numerical values, not an event in the ticket sample space.

Inverse image: outcomes that produce selected values

For $D \subseteq \mathbb{R}$, the inverse image is

$$X_B^{-1}(D) = \{t \in \Omega : X_B(t) \in D\}.$$

In particular,

$$X_B^{-1}(\{1\}) = \{t_1, t_2, t_5\} = B,$$

and

$$X_B^{-1}(\{0\}) = \{t_3, t_4, t_6\} = B^c.$$

An inverse image is a set of outcomes; it need not be an inverse function. The inverse image connects a numerical condition on a random variable to an event to which the probability measure applies:

$$P(X_B = 1) = P(X_B^{-1}(\{1\})) = P(B) = \frac{3}{6} = \frac{1}{2}.$$

The distribution of X_B is $P(X_B = 1) = 1/2$ and $P(X_B = 0) = 1/2$. The function X_B , its inverse-image event B , and the number $P(B)$ are three different objects.

Checkpoint 5. Find $X_B(\Omega)$, $X_B(A)$ and $X_B^{-1}(\{1\})$. Explain which are sets of real values, which are sets of ticket outcomes, and which is a probability.

7. A continuous example: ticket closure time

The triplet is often not written out in routine probability calculations. To see what remains implicit, consider a ticket selected from a specified service process, and let T_c be its closure time, measured in hours. The question

What is the probability that this ticket takes more than one day to close?

is written $P(T_c > 24)$, since one day is 24 hours. In an underlying ticket-system experiment, each outcome ω determines a closure time $T_c(\omega)$ in hours. The event of interest is the inverse image

$$T_c^{-1}((24, \infty)) = \{\omega \in \Omega : T_c(\omega) > 24\}.$$

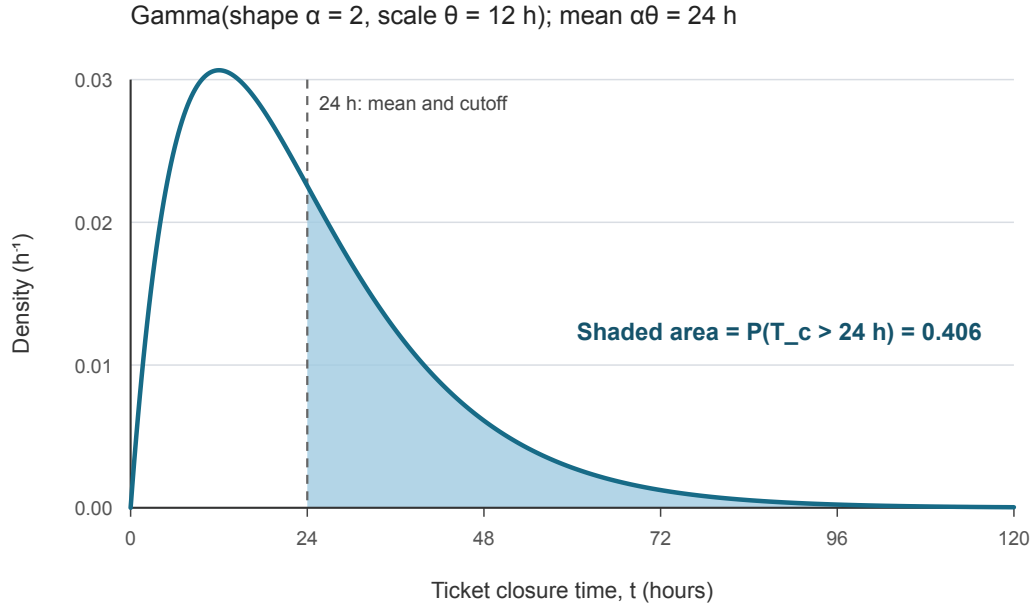
For an illustrative calculation, suppose the closure time follows a Gamma distribution with shape $\alpha = 2$ and scale $\theta = 12$ hours. These are hypothetical model parameters, not estimates from the six-ticket audit. Under the shape-scale parameterization, the density is

$$f_{T_c}(t) = \frac{t^{\alpha-1} e^{-t/\theta}}{\Gamma(\alpha) \theta^\alpha}, \quad t \geq 0.$$

With $\alpha = 2$ and $\theta = 12$ hours, this becomes

$$f_{T_c}(t) = \frac{t}{12^2} e^{-t/12}, \quad t \geq 0,$$

where t is measured in hours. Here α is the shape parameter, not the mean. The mean in this shape-scale parameterization is $E[T_c] = \alpha\theta = 24$ hours. The plotted curve shows this model and shades the area to the right of the 24-hour cutoff, which coincides with the mean.



Gamma density for ticket closure time T_c under the illustrative model. The vertical line marks both the mean and the 24-hour cutoff; the shaded right tail represents $P(T_c > 24 \text{ h})$.

For the six-ticket experiment above, probabilities were calculated by counting equally likely IDs: $P(C) = |C|/6$. A continuous ticket time cannot be handled by counting individual possible durations. Under a density model, the probability of an interval is the area under the density over that interval. A density value is not itself a probability; integration accumulates the density across the range. In the audit model, the corresponding finite calculation is a sum of singleton probabilities; for example, $P(C) = \sum_{t_i \in C} P(\{t_i\}) = 4/6$. The closure-time probability is the integral of its density over the event. With the substitution $u = t/12$,

$$\begin{aligned} P(T_c > 24) &= \int_{24}^{\infty} \frac{t}{144} e^{-t/12} dt \\ &= \int_2^{\infty} u e^{-u} du \\ &= (2 + 1)e^{-2} = 3e^{-2} \approx 0.406. \end{aligned}$$

Thus, under this stipulated model, the probability that closure takes more than 24 hours is about 0.406. A different shape, scale or target ticket population would give a different value. Specifying the Gamma family alone would not suffice; the parameters and their parameterization are needed.

The triplet for the closure-time model

For this calculation, use the following canonical probability space, which models closure time alone:

$(\Omega_c, \mathcal{F}_c, P_c)$.

In the calculation above, the shorthand $P(T_c > 24)$ denotes $P_c(T_c > 24)$ on this space. It is distinct from P , the uniform measure used for selecting one of the six audited ticket IDs.

1. **Sample space:** $\Omega_c = [0, \infty)$, the possible closure durations measured in hours.
2. **Event collection:** $\mathcal{F}_c = \mathcal{B}([0, \infty))$, the Borel subsets of the nonnegative real line.
3. **Probability measure:** for every $E \in \mathcal{F}_c$,

$$P_c(E) = \int_E f_{T_c}(t) dt, \quad f_{T_c}(t) = \frac{t}{144} e^{-t/12}, \quad t \geq 0.$$

The density integrates to one, so $P_c(\Omega_c) = 1$. Define the random variable by $T_c: \Omega_c \rightarrow [0, \infty)$, $T_c(t) = t$. Thus the event of closure after 24 hours is

$$T_c^{-1}((24, \infty)) = (24, \infty) \in \mathcal{F}_c,$$

and its probability is

$$P_c(T_c > 24) = P_c((24, \infty)) = \int_{24}^{\infty} f_{T_c}(t) dt.$$

This is one explicit probability space realizing the assumed Gamma law. A full ticket-system model could use outcomes that also include ticket type, priority and other information, with closure time as a function on that larger space. In routine calculations, $T_c \sim \text{Gamma}(2, 12)$ states the induced law needed for probabilities about closure time without writing the whole triplet. The distribution of T_c does not by itself describe every other random quantity in the ticket system.

This is why many engineering and AI calculations work directly with random variables and their distributions after the probability-space foundation has been established.

8. Worked problems

Problem 1: probability of “Other”

Find the probability that a uniformly selected ticket has neither Billing nor Account Support required according to the reviewer.

Solution. By definition, $O = (B \cup A)^c = \{t_4\}$. Therefore

$$P(O) = \frac{1}{6}.$$

Problem 2: exactly one named destination is required

Find the event and probability that exactly one of Billing and Account Support is required.

Solution. The event is

$$(B \setminus A) \cup (A \setminus B) = \{t_2, t_3, t_5, t_6\}.$$

Its probability is $4/6 = 2/3$.

Problem 3: agreement between reviewer and model

Let $M(t)$ be the model's classification and $R(t)$ the reviewer's reference classification for ticket t . Define exact agreement as equality of these classification sets. The agreement event from the table is

$$C = \{t_2, t_3, t_5, t_6\}.$$

Under uniform selection, $P(C) = 4/6 = 2/3$. This equals the observed agreement fraction because the experiment selects uniformly from this exact audit.

Problem 4: use an inverse image

Define $Y(t) = 1$ if the model classification exactly matches the reviewer and $Y(t) = 0$ otherwise. Then $Y^{-1}(\{1\}) = C$, and

$$P(Y = 1) = P(C) = \frac{4}{6} = \frac{2}{3}.$$

The random variable records an outcome numerically; its inverse image gives the event; the probability measure assigns that event a number.

9. Connection to AI evaluation and continuation

The probability $2/3$ here concerns uniform selection from six audited ticket IDs. It is not automatically the model's agreement probability on future traffic. That requires a defined target population, an appropriate sampling process and assumptions connecting this audit to future tickets. Changing the reviewer protocol may also change the event called “agreement.”

The same distinctions arise in AI evaluation. A test-set accuracy is an observed proportion. Interpreting it as a probability of correct classification for future cases requires additional assumptions. If model generation is stochastic, its repeated-output distribution is another probability model with a different outcome space. In each case, specify the outcomes, event and sampling

or generation process before interpreting a probability.

A suitable stopping point is the triplet and the calculation of $P(B \cup A) = 5/6$. Continue to the sigma-algebra and axioms once the class can identify the complement event O . The random-variable and image sections can continue in a later meeting. Bertrand's chord problem is posed here but not derived; the rigorous supplement specifies its sampling models and works through their parametrizations and probabilities.

References

- Jay L. Devore, *Probability and Statistics for Engineering and the Sciences*. Chapter 2, "Probability and Random Variables," as identified by the instructor. Edition and page references should be checked against the course copy.
- Krishna Jagannathan, *Probability Foundations for Electrical Engineers*, lecture notes, 9 July 2025: Chapter 1 (sets and functions), Chapter 4 (probability models), Chapter 5 (probability rules), Chapter 6 (discrete models), and Chapter 11, Definitions 68–70 (measurable functions and laws).
- Alan Hájek, "Interpretations of Probability," *Stanford Encyclopedia of Philosophy*, comparative overview: <https://plato.stanford.edu/entries/probability-interpret/>.
- John D. Norton, "Bertrand's Geometrical Paradoxes: Proofs," University of Pittsburgh, presenting the chord constructions yielding $1/3$, $1/2$ and $1/4$: https://sites.pitt.edu/~jdnorton/teaching/paradox/chapters/probability_for_indifference/Bertrand_proof/Bertrand_proof.html.

The supplement supports the same notation and gives the technical detail. All six-ticket data and classifications are hypothetical.

All numerical values are hypothetical and stipulated for teaching.